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Electroless plating of iron-group metals and electrochemical comparison for thermoelectric contacts

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ABSTRACT

Bismuth telluride and lead telluride are typical thermoelectric semiconductors that are metallized for electrical contacts before being assembled into devices. In this study, three iron-group metals of nickel cobalt iron, were chemically plated on n-type bismuth telluride as the prospective contact material for adhesion and barrier in soldering connection with copper. Aluminum was used as the sacrificial anode to ensure iron deposition. The sensitization in borohydride solution facilitated the subsequent plating of metals. The plating rates were determined by gravimetric method. The crystalline structure and morphology of films were characterized. Electrochemical polarizations were undertaken to compare the driving force among three plating reactions through the reduction of metallic cations and the oxidation of hypophosphite anions with the mixed potential theory. The heterojunction of bismuth telluride with copper was fabricated by reflow soldering with $Sn_{95}Sb_5$ foil, and the interfacial resistivity was evaluated by multiple resistance measurements across the junction.

1. Introduction

The energy conversion between heat and electricity can be accomplished directly in an all-solid-state way by thermoelectric materials via Seebeck and Peltier effects. Metal tellurides are important thermoelectric semiconductors either in refrigeration or power generation [1,2]. Bismuth telluride compounds have been used practically in semiconductor cooling near room temperature [1]. Lead telluride compounds represent one class of semiconductors for converting thermal energy into electricity at high temperature [2]. Over the past decades, the thermoelectric performance of PbTe-based materials has been greatly enhanced by the state-of-the-art material fabrication and characterization techniques in a synergistic strategy [2]. As an example, the sodium doped *p*-type PbTe and the iodine doped *n*-type PbTe materials, have been witnessed with a peak *ZT* in excess of 1.8 and 1.4, respectively, which promises an energy efficiency of 12 % [3].

In view of the thermoelectric device the contact resistance between thermoelectric semiconductors and metal electrodes, is desired to be as low as possible so that the parasitic power loss is minimized and the energy efficiency enhanced [4]. For this purpose, a special type of metal is usually deposited on the semiconductors prior to the soldering connection or direct bonding to metal electrodes [4]. For example, in the Peltier cooling devices, a thin layer of nickel on bismuth telluride has been utilized effectively as an adhesion for wetting solder and a barrier for inhibiting unwanted atomic diffusion [5]. In addition, nickel is considered an excellent interface material with low contact resistivity in the electronic packaging applications, such as in printed circuit boards and solar cells [6–9]. Nickel is also resistive to corrosion and wear for surface engineering application. Nickel film can be formed either by physical sputtering deposition or by chemical deposition through an electroplating or electroless plating process.

Although bismuth telluride devices have been developed for thermoelectric generators by harvesting electricity from low-grade heat [10,11], lead telluride devices are seldom demonstrated in prototyping electric power output from high-temperature heat source. One of the obstacles in the development of such devices is the electrical interconnection between the n- and *p*-type PbTe with metal electrodes [3], which is expected to withstand up to 600 °C at the hot side and maintain an interfacial resistivity in an order of magnitude of 10^{-9} Ω m^2. Such harsh temperature poses a huge challenge to the electrical and mechanical reliability of the metal to semiconductor connection for long-term service [4]. The practical output electricity and energy conversion efficiency are discounted by the contact resistance in the devices and the parasitic thermal loss in the generator system. In particular, iron has

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been chosen as the electrical contact material or metal electrodes in lead telluride thermoelectric generators [12–15]. For example, under uniaxial pressure, a pure iron electrode was combined with PbTe bulk by co-firing SnTe or Fe powder as a buffer layer [16,17]. The contact resistivity was reported less than 10 μΩ cm^2 at the metallurgical junction [16]. The mismatch of thermal expansion coefficients between iron electrodes and PbTe could induce thermal stress and even crack at the interface.

An alloy of iron and cobalt ($Co_{80}Fe_{20}$) has been investigated as the contact material for lead telluride and other chalcogenides [3,18]. The metallization was realized by sintering a layer of alloy powders with PbTe bulk or with the compacted PbTe powders together. The contact resistivity was reported at the level of 10~20 μΩ cm^2 at the ambient temperature [3]. However, the sintering metallization has to be carried out in a hot-pressing furnace, and it is not easy to achieve an even thickness and mass-production. Recently, cobalt was studied as the potential contact material for PbTe semiconductor [19–22]. In contrast, the Co film was formed by an electroless chemical reduction of cobalt ions in solution. The contact resistivity of Co with *n*-type PbTe is 21 μΩ cm^2, and 27 μΩ cm^2 with *p*-type [21].

In this study, as an instance we plate Ni, Co, and especially Fe, the same group of transition metal, onto *n*-type semiconductor of bismuth telluride by an electroless chemical plating method. It is deemed that cobalt and iron are the suitable candidate as the contact material in the bismuth telluride device for power generation, since they have worked well at high temperature in the lead telluride device [3,12,19]. A general methodology was developed for chemical metallization of the thermoelectric telluride in an autoclave. The electrochemical mechanism was studied with the mixed potential theory by understanding polarization kinetics of the redox process [23,24]. Finally, $Sn_{95}Sb_5$ solder was used to connect the Fe–Co plated bismuth telluride with copper, and the electric contact at the interface was evaluated.

2. Experiments and methods

Bismuth telluride samples, 4 mm × 4 mm × 10 mm of rectangular bar and Φ12.6 mm × 2 mm of circular disc, were made by wire cutting polycrystalline ingot. The chemical composition of the ingot is $Bi_2Te_{2.7}Se_{0.3}$, which is *n*-type semiconductor. The bar-shape samples were used for the chemical plating and then the soldering purposes, while the discs for electrochemical analysis as working electrode. The samples were cleaned ultrasonically in ethanol for 5 min, in an alkaline solution for 5 min, in an acetic solution for 5 min, and rinsed with deionized water.

2.1. Electroless chemical plating of iron-group metals on bismuth telluride

All the chemicals (A.R. grade) were purchased from Aladdin or Sinopharm Chemical Reagent Co. in China. $NiSO_4{\cdot}6H_2O$, $CoSO_4{\cdot}6H_2O$ and $Fe(NH_4)_2{\cdot}(SO_4)_2{\cdot}6H_2O$, with an abbreviation of $MeSO_4$, were used as the nickel, cobalt and iron sources for electroless chemical plating on bismuth telluride, respectively. Sodium citrate ($Na_3C_6H_5O_7{\cdot}2H_2O$) and/or citric acid ($H_3C_6H_5O_7$) were used as the complexing agents, and sodium hypophosphite ($NaH_2PO_2{\cdot}H_2O$) or potassium borohydride (KBH_4) as the reducing agents. Table 1 gives the chemicals in the four baths and the conditions for plating the three metals. Each formulation was an aqueous solution of 50 mL, and the pH was tuned with citric acid or ammonia solution.

Table 1
Solution chemicals and conditions for plating the three metals.

Chemicals & conditions	Ni-1 (acid)	Ni-2 (base)	Co (base)	Fe (base)
$MeSO_4$ (mM)	90	90	65	50
NaH_2PO_2 (mM)	370	370	283	340
$Na_3C_6H_5O_7$ (mM)	68	68	102	136
NH_4Cl (mM)	–	224	280	486
pH	3–4	9–10	9–10	10
Temperature (°C)	80–85	90–95	90–95	70–80
Time (hour)	4–6	6–8	6–8	3–6

The bar samples were dipped in an alkaline borohydride solution for sensitization 5 s, and dipped in one of the plating baths for activation 5 s. Then the plating experiments were performed in a Teflon-liner stainless-steel autoclave with 100 mL in volume. The bars were suspended in the plating baths either by a cotton thread for plating nickel and cobalt or by an aluminum wire (0.8 mm in diameter) for plating iron. The autoclave was sealed for the heating treatment. Afterwards, the plated samples were rinsed with water, and stored in ethanol for next usage.

2.2. Characterization of the metal films

The weight of each telluride sample, after plating for various hours, was recorded by an electronic balance with 0.1 mg precision. The gravimetrical data and the sample geometry were adopted to calculate the film thickness over time and the plating rate for each metal. The crystalline structures of the films were characterized by X-ray diffractometer (XRD; D8-ADVANCE) with Cu Kα radiation as the incident beam and six degrees per minute under 40 kV and 30 mA. The surface morphology of the films was imaged by scanning electron microscope (SEM; QUANTA 450 FEG) at 20 kV, and the energy dispersive spectroscopy (EDS) was used for element analysis. The cross section of the films and the subsequent soldering junction were examined on the polished surface.

2.3. Electrochemical analysis of the electroless chemical plating mechanism

An electrochemical workstation (CHI660E, CH Instruments, China) was employed to investigate the plating mechanism in an electrolyte cell of 100 mL. The working electrode was bismuth telluride disc, which was accommodated in a Teflon mould and exposed with a circular area of 1.0 cm^2 to the electrolyte solution for surface reactions. Bismuth telluride was polished and cleaned prior to each measurement. Saturated calomel electrode (SCE) was used as the reference, platinum plate with 1cm × 1 cm as the counter electrode. The electrolyte solutions were similar to the plating baths in the chemicals, except the absence of metallic salts in the anodic electrolytes and the absence of the reducer in the cathodic electrolytes. After the open-circuit potentials (OCPs) were stabilized in the anodic or cathodic electrolyte solutions, polarization curves were measured by applying a dynamic potential (5 mV/s) on the electrode around the OCPs in the positive (anodic) or negative (cathodic) way, respectively. In particular, in the anodic reaction of the reductant for the iron-plating electrolyte, the working electrode was also connected with an aluminum wire in the electrolyte solution for comparison. The polarization tests were undertaken both at the ambient and plating temperatures. The temperature of the electrolyte cell was raised and maintained in a hot-water container.

2.4. Fabrication of the heterojunction with copper and the interfacial resistivity

The four lateral planes of the semiconductor bar (bismuth telluride, 4 × 4 × 10 mm^3), were covered by using photoresist followed by a UV curing process. The two end planes were plated with iron and cobalt film in two steps using the recipes in 2.1, and then the photoresist was dissolved chemically. The heterojunction between the plated semiconductor and copper bar of the same size, was fabricated by reflow soldering from one end to the other in nitrogen flow. A graphite mould was used for the soldering connection. The soldering material was $Sn_{95}Sb_5$ foil with 0.1 mm thickness.

The electrical resistance across the interface was measured by an instrument (ZEM3, Ultra-Riko, Japan) in the four-probe configuration [25]. As illustrated in Fig. 1, the two voltage probes with 5.9 mm in gap

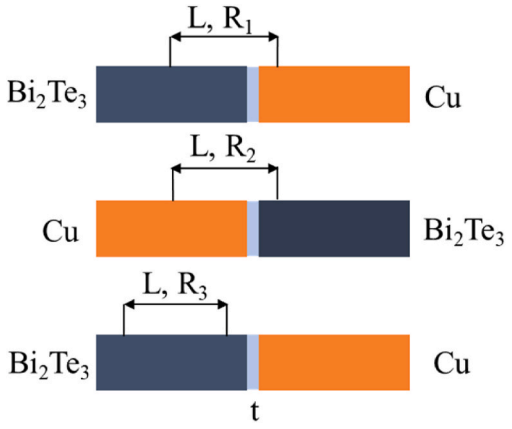


Fig. 1. Schematic configuration for measuring interface resistance of the heterojunction.

(L), were asymmetrical with respect to the interface. Hence, the heterojunction could be placed in two ways, i.e., being upright (Bi_2Te_3–Cu) and upside down (Cu–Bi_2Te_3), and two resistances (R_1, R_2) were measured. The resistance across bismuth telluride only (R_3) was measured as well. The copper resistance is negligible. The interfacial resistivity could be calculated from the three resistances and the interface thickness (t).

3. Results and discussion

3.1. Microstructure and surface morphology of the plating metals

The XRD patterns of four metal films are shown in Fig. 2. The Ni films are plated either from the acidic or alkaline baths (Table 1). The wide peak for the Ni films is characteristic of the amorphous Ni–P phase [26]. The peak position is consistent with the strongest peak of Ni(111) plane if nickel crystallizes in the FCC structure (JCPDS card no. 4–856). The four peaks of (121) (100) (002) (101) for the Co film are attributed to the diffraction from the hexagonal phase (HCP structure, JCPDS card no. 5–727) of cobalt. The low intensity indicates that cobalt film does not crystallize well. In contrast, the iron film displays two sharp peaks that correspond to the (110) and (200) crystalline planes of α-Fe (BCC structure, JCPDS card no. 6–696). The telluride substrates are covered by a thick layer of the plated metal so that there is no any diffraction peak from the telluride.

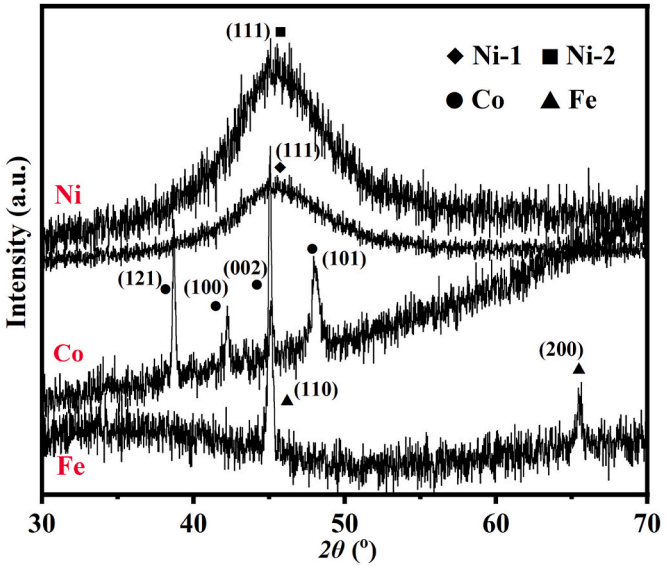


Fig. 2. The XRD patterns of the three metal films plated on n-type bismuth telluride.

The SEM images of surface morphologies of four films and the corresponding EDS elemental analysis are presented in Fig. 3. The nickel and cobalt films display similar topologies that consist of round grains with tens of micrometers in size. The grain boundaries can be distinguished. The large grains seem to be agglomerated from smaller ones. The similar morphology reflects the similar plating mechanism for Ni and Co film. The Co film appears more rough on the surface. In contrast, the crystallites of the iron film exhibit sharp edges. Three edges share one common vertex for each grain. It can be seen that the angle is **90°** between the adjacent edges, which is in agreement with the cubic structure of α-Fe. The iron microcrystals are grown with the [111] crystallographic axis being normal to the substrate. The crystalline grains are less than 10 μm in size, and they are coalesced with each other and produce the roughness of film. Some vertexes of the grains are abrased away or corroded with a dark spot, possibly when the sample is handled by tweeze. The EDS elemental spectra show the films are not pure in the chemical compositions, and they contain phosphorus as an impurity. The phosphorous amount differs among the three kinds of films. The high phosphorous content makes nickel be amorphous in structure, no matter how it is plated in the acidic or alkaline bath. The middle amount of phosphorus leads to poor or nanoscale crystallization of cobalt. Due to the low percentage of phosphorus, the iron film crystallizes well. The large interstitial site in the α-Fe BCC structure favors the accommodation of impurity atom as well.

3.2. Chemical plating rates and the redox reactions

Prior to the continuous plating experiments, bismuth telluride samples are treated by the sensitization and activation so that a seeding layer of each metal is formed at first. In specific, it is sensitized by surface wetting in an alkaline KBH_4 solution, and subsequently activated in each of the plating baths. The process is completed in 10 s. Afterwards, a yellowish thin film is created.

Table 1 presents the chemical formulations of the electroless plating baths and the process parameters for the continuous deposition of three kinds of metals onto *n*-type Bi_2Te_3 samples, in which NaH_2PO_2 is the reductant rather than KBH_4. Fig. 4 displays the weight increments in relation to time for the three alkaline-plated metals. Overall, the weight of each film increases monotonously with the plating time, and the relationships are approximately linear. This indicates an autocatalytic mechanism can be accounted for plating metals, since the atomic depositions occur continuously and no external current is passed to the substrate. It also implies that the metal cations and the reducing anions in each bath are enough in amount for the duration of the film deposition. The film thickness is calculated from the weight increments and metal density. The fitting slope in Fig. 4 represents the plating rate. The plating rates are close to each other between nickel and cobalt, being 6–7 μm/h. The plating rate of iron is lower at 2.6 μm/h. It could be noted that the standard reduction potentials for Ni^{2+}/Ni, Co^{2+}/Co, and Fe^{2+}/Fe pairs, are −0.257V, −0.280 V, and −0.447 V, respectively. The more positive the potential is, the more noble the metal is, and the easier the metallic cation would be reduced.

Aluminum wire works as a sacrificial anode for the cathodic reduction of ferrous cations [27], otherwise, the iron deposition does not happen by use of hypophosphite only. As shown in Fig. 4, the deposition of Ni and Co does not occur apparently in the first hour of reaction, because the plating solutions in the autoclave are not heated to the setting temperature. However, the plating of iron takes place in 1 h, though the reduction potential of Fe^{2+}/Fe is more negative. Therefore, the plating of iron is promoted by the galvanic dissolution of the contacted aluminum wire even at the low temperature. As the aluminum is dissolved into the alkaline solution, electrons are released and transferred to the telluride surface for helping the iron reduction. The Al wire

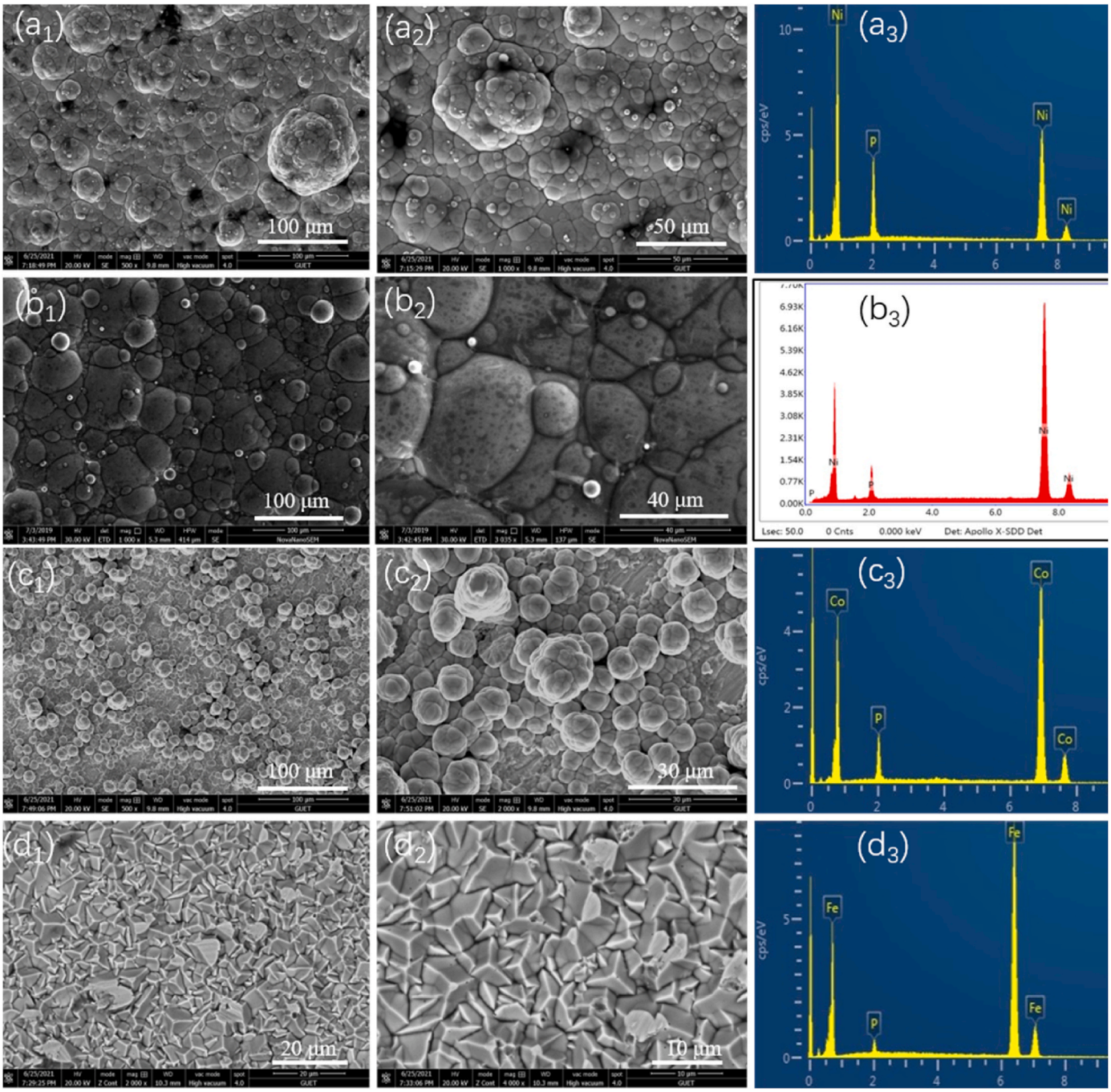


Fig. 3. The SEM images and EDS elemental spectra for the acidic Ni plating (a_1-a_3), the alkaline Ni plating (b_1-b_3), the alkaline Co plating (c_1-c_3) and the alkaline Fe plating (d_1-d_3) on n-type bismuth telluride.

does become thin in the diameter of the part immersed in the solution after the plating experiment. In comparison, in the alkaline ferrous bath without hypophosphite, the iron does not deposit owing to the connected aluminum. It shows a synergistic effect between hypophosphite anions and aluminum metal in the iron plating process. It resembles the case that the galvanic corrosion between two dissimilar metals does not occur in an electrolyte without any oxidizer. Herein the electroless plating of metal could be considered as the reverse of the electrochemical corrosion.

In the acidic bath of plating Ni, the dominant redox reaction can be described in Reaction (1), whereas the side reaction is the disproportionation of hypophosphorous acid in Reaction (2). Ni^{2+} is present in the form of $[Ni(Cit)_x]^{2-3x}$ complex ions, where Cit represents the ligand of citric anion. In addition, hydrogen evolution reaction occurs readily in the acidic bath, since the hydrogen ions could be reduced by the hypophosphite with hydrogen gas being released. It agrees with the phenomenon that some gas bubbles are found absorbed on the inner wall after the plating experiment. In contrast, in the alkaline solutions for depositing the iron-group metals (Ni, Co, Fe), Reaction (3) is the predominant process that is accompanied by the disproportionation in Reaction (4). Here Me^{2+} denotes one of the three metal cations, and $[Me(Cit)_x]^{2-3x}$ is the complex ions. No gas bubble on the wall supports that the hydrogen evolution reaction is inhibited in the high-pH baths. However, the tiny hydrogen bubbles are possibly evolved on the surface of bismuth telluride, and they work as the nucleation sites for nickel or cobalt deposition, so that the grains tend to be spherical in Fig. 3.

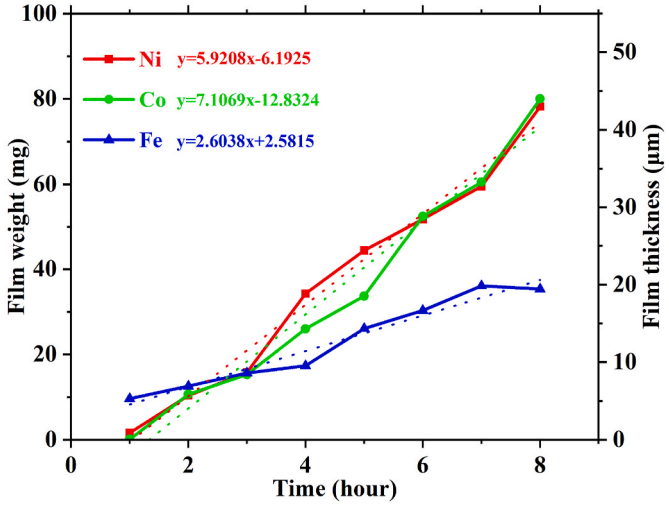


Fig. 4. The film weight and thickness of the three metals as functions of the plating time.

Furthermore, the presence of hydroxide ions in the alkaline bath has retarded Reaction (4), and as a result less amount of phosphorous has be produced and incorporated into the metal films. This is consistent with the EDS elemental results in Fig. 3 that the phosphorous content is lower in the alkaline plated Ni than in the acidic plated Ni. Large amount of phosphorus atoms are deposited concurrently with Ni owing to the disproportionation reaction, and the resultant Ni–P film become amorphous. The disproportionation is affected by the chemicals in the baths.

$$[NiCit_x]^{2-3x} + H_3PO_2 + H_2O = Ni + xCit^{3-} + H_3PO_3 + 2H^+ \quad (1)$$

$$3H_3PO_2 = H_3PO_3 + 2P + 3H_2O \quad (2)$$

$$[MeCit_x]^{2-3x} + H_2PO_2^- + 3OH^- = Me + xCit^{3-} + 2H_2O + HPO_3^{2-} \quad (3)$$

$$3H_2PO_2^- = HPO_3^{2-} + 2P + OH^- + 2H_2O \quad (4)$$

3.3. Polarization curves of the half reactions

The electroless chemical plating process can be separated into two electrochemical half reactions occurring simultaneously on bismuth telluride, which are the reduction of metal cations and the oxidation of the reductant. The electron transfer in the redox reactions takes place via the semiconductor surface. Electrochemical polarization test is intended to induce the cathodic or anodic half reaction by an applied dynamic potential on the working electrode versus the open-circuit potential [28].

Fig. 5 displays the anodic polarization curve for the potassium borohydride in the anodic solution (sensitization) and the cathodic polarization curves for the three metal cations in the cathodic solution (activation) at ambient temperature, respectively. The electrolyte solutions are adjusted to pH 8–9 by ammonia solution. The polarization tests are initiated from the open-circuit potentials, and the potentials are changed linearly either in the positive (anodic) or negative (cathodic) direction, respectively. The open-circuit potentials of bismuth telluride in the citrate solutions of Ni(II) and Co(II) salt are close to each other at 80 mV, while the potential in the citrate solution of Fe(II) salt is at −50mV. It indicates that the Co(II) and Ni(II) have the similar tendency towards reduction, and it is harder to get the Fe(II) ions reduced. In contrast, the open-circuit potential of bismuth telluride in the BH_4^- ammonia solution is at −620mV. Thus, the equilibrium potential of bismuth telluride in the sensitization solution is much more negative than those in the activation solutions. The potential difference is as large as 570 mV for Fe(II) and 700 mV for Ni(II) and Co(II), respectively. It

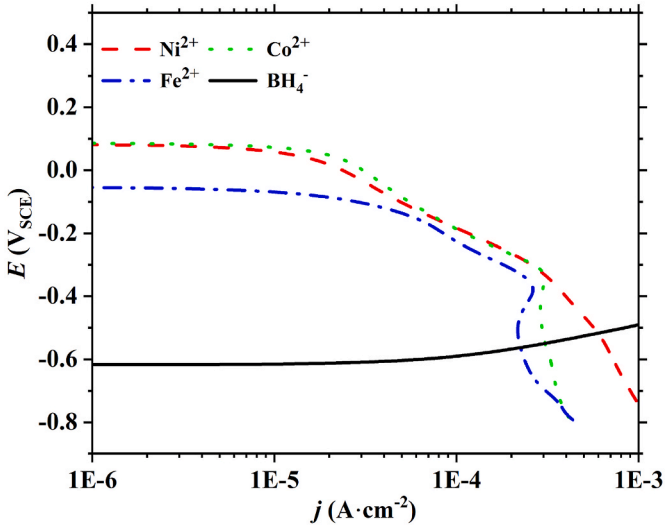


Fig. 5. The anodic polarization curve for the borohydride anion and the cathodic polarization curves for the three metal cations at ambient temperature at pH 8–9.

exhibits a large driving force for the redox reactions between the reductant of BH_4^- anion and the three metal cations. The driving force ensures that the metal can be deposited easily onto the surface when the sensitized sample, which is soaked with the BH_4^- electrolyte, is immersed into the activated solution. This is consistent with the phenomena that a thin layer of yellowish metal is formed on the sample after the immersion for 5 s. In Fig. 5, the intersections between the anodic branch and the three cathodic branches represent the mixed potentials and depositing currents for the three metal deposits [29]. However, if the anodic and cathodic solutions are mixed together, the redox reactions are vigorous, and large amount of black metal particles are precipitated out of the solution.

Fig. 6 presents the cathodic polarization curves in the three cathodic solutions of metal cations and the anodic polarization curve in the anodic solution of sodium hypophosphite at ambient temperature, respectively. The electrolyte solutions are maintained at pH 10–11, which is consistent with the plating baths. As the open-circuit potential of the telluride in the NaH_2PO_2 solution is more positive than those in

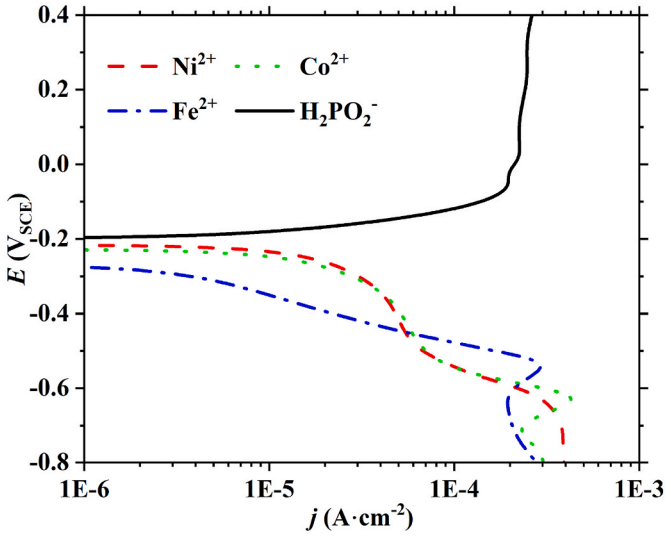


Fig. 6. The anodic polarization curve for the hypophosphite anion and the cathodic polarization curves for the three metal cations at ambient temperature at pH 10–11.

the metallic cation solutions, it is reasonable that the NaH_2PO_2 is not capable of reacting with the metallic cations at ambient temperature. The reducing ability of NaH_2PO_2 is less than that of the KBH_4. For the anodic polarization, the anodic current increases abruptly at the beginning owing to the oxidation of $H_2PO_2^-$ anions, and the current becomes saturated when the potential increases further, which shows the electrochemical oxidation reaction is limited by the ion diffusion process. For the cathodic polarizations, the metal reductions are influenced both by the electron transfer and ion diffusion process.

In comparison, Fig. 7 illustrates the polarization curves at 85 °C in the same electrolyte solutions, and all the potentials are lowered as compared with those in Fig. 6. In Fig. 7 the equilibrium potential in the NaH_2PO_2 solution is less than those in the Ni^{2+} or Co^{2+} containing solutions. Therefore, the two cations could be reduced by the NaH_2PO_2 at 85 °C. This is consistent with the plating experiments that the Ni and Co metal are deposited onto the telluride surface. As the driving forces for the redox reaction with NaH_2PO_2 are not as large as those for the reaction with KBH_4, the plating rates are mild for Ni and Co. For example, it will take approximate 4 h to grow 20 μm in thickness, and in the meanwhile the bath is stable and no black metal particles are precipitated. That is to say, the homogeneous nucleation of Ni and Co does not occur in the plating baths. In contrast, the equilibrium potential in the Fe^{2+} containing solution is more negative than that in the NaH_2PO_2 solution so that the ferrous cation cannot be reduced by the $H_2PO_2^-$ anion. Consequently, no iron film is formed in the plating experiment with the NaH_2PO_2 as the reducing agent. However, if the aluminum wire is kept in contact with the working electrode, the open-circuit potential in the NaH_2PO_2 solution is much lowered to −680 mV in Fig. 7. In this way the aluminum wire would work like a secondary reductant. The electrons for the ferrous reduction are contributed not only from the NaH_2PO_2 oxidation but also from the oxidation of aluminum. In the plating process, as the aluminum is dissolved into the alkaline solution, the electrons are transferred to the telluride surface, where the ferrous ions accept the electrons and are transformed into iron atoms for attachment on the surface. Therefore, the phosphorus content is low in the iron film.

3.4. Microstructure and interfacial resistivity of the contact

Fig. 8 is the cross-section SEM images of the contact between the two-step metallized bismuth telluride and copper, which are soldered by reflowing $Sn_{95}Sb_5$ foil. The interfacial region of interest consists of

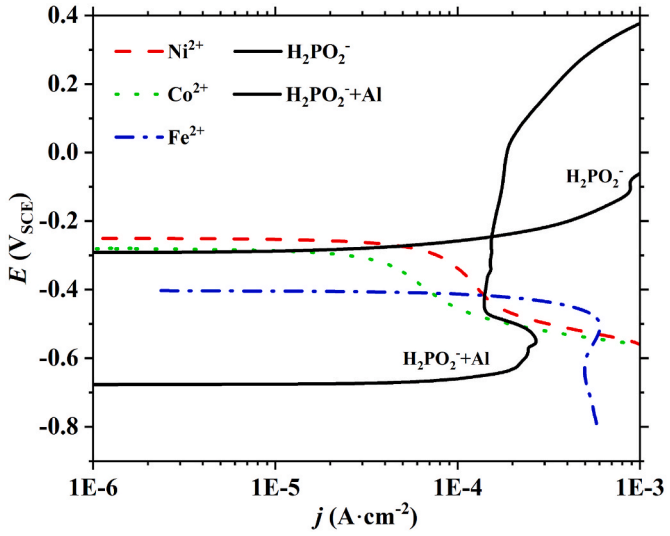


Fig. 7. The anodic polarization curves for the hypophosphite anion with the electrode in contact with Al, and the cathodic polarization curves for the three metal cations at 85 °C at pH 10–11.

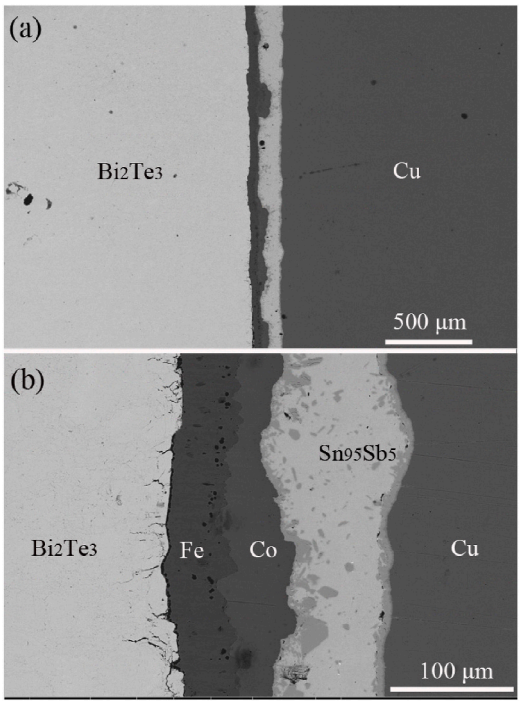


Fig. 8. The cross-section SEM images of the interface region between bismuth telluride and copper, which consists of the chemically plated iron and cobalt bilayer and the Sn–Sb layer.

bismuth telluride, iron, cobalt, solder and copper from left to right. The two edges of bismuth telluride and copper are in parallel approximately, nevertheless the border is not straight between the plating metal and solder. It is examined that the iron layer is uniform in thickness and conformal along the interface but the cobalt layer is not. This is consistent with the morphological overview that the cobalt film appears rough owing to large spherical grains in Fig. 3. Due to the brittleness, some cracks are developed in bismuth telluride near the interface in the specimen preparation. Since iron is active in water, and in the polishing process it is electrochemically eroded by water as an anode while the bismuth telluride and other materials work as a large cathode [30]. The iron at the interface is corroded locally owing to the easy electron transfer there and the possible stress, which is caused by the mismatch of thermal expansion coefficient between semiconductor and metal. As the iron on the surface is etched along the interface, the abrasion action induces the surface crack of the brittle n-type bismuth telluride. A distinct layer of intermetallic compound (IMC) is produced between the copper and solder, which is usually Cu_3Sn phase [31]. A thin layer of IMC is beneficial to the joint strength, while a thick layer would be harmful to the electrical transport. In contrast, the intermetallic compound across the cobalt and solder is presented deep into the solder matrix. By the proposed method [25] and the modification in Fig. 1, the interfacial resistivity across the bismuth telluride and copper is determined to be 73 μΩ cm^2 at 25 °C and 99 μΩ cm^2 at 110 °C. The resistivity is larger than what is desired for making thermoelectric devices, which is preferred to be less than 10 μΩ cm^2. The relative large resistivity is attributed to oxidation of the specimen, since a small amount of air is trapped in the graphite mould. The resistivity at higher temperatures and the barrier performance of the iron-group alloy films are under investigation by improving the soldering process.

4. Conclusion

The iron-group metals of nickel cobalt iron are plated on bismuth telluride respectively in the citrate−metallic electrolyte baths at 70–90 °C. Sodium hypophosphite is a common reductant. Aluminum

promotes the iron deposition as a sacrificial anode. The sensitization in the alkaline KBH_4 solution is favorable for seeding metal and ensuring the subsequent deposition in the plating baths. The plating rate is 6~7 μm in thickness per hour for nickel and cobalt, and 2.6 μm/h for iron. Phosphorus is incorporated in the metal films, among which the Ni is amorphous and Fe is crystalline. The Ni and Co films consist of agglomerated round grains, whereas the Fe film displays cubic crystallites. The electrochemical mechanism of the plating process is presented with the driving force of the redox reaction by the cathodic and anodic polarizations. The metallurgical junction between copper and the telluride semiconductor, which is metallized with Fe–Co bilayer, is examined on the microstructure, and the interfacial resistivity is larger than what is expected for making thermoelectric devices. Other metallization strategies would be attempted to reduce the resistivity, such as electroless plating of Co–Fe bilayer or alloy onto the thermoelectric telluride. The proposed plating method could be extended to the metallization of other metal telluride and chalcogenide thermoelectric semiconductors.

Author declarations

The authors have no conflicts of interest to disclose.

CRediT authorship contribution statement

Liming Zhu: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Dezhu Sun:** Methodology, Investigation, Formal analysis. **Xuyang Li:** Visualization. **Wei Liu:** Formal analysis. **Jiaqiang Huang:** Formal analysis, Funding acquisition. **Caihang Liang:** Supervision. **Xiaokai Hu:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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